

EduQuest: An NLP-Powered Semantic Course Recommender for NUS Students

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Abstract

Course selection is a critical and often overwhelming process for NUS students, particularly when academic or career interests are difficult to express through simple keywords. NUSMods, the primary tool used for module exploration, operates using literal keyword matching, which does not capture semantic intent. As a result, students frequently browse inefficiently, rely on hearsay, or overlook modules that align with their deeper goals.

This project introduces **EduQuest**, an NLP-powered semantic course recommender that accepts open-ended free-text queries and returns personalized module suggestions grounded in semantic similarity rather than keyword overlap. EduQuest employs a two-stage pipeline: (1) query enrichment using a large language model (LLM) to transform vague student queries into detailed pseudo course descriptions, and (2) embedding-based retrieval using cosine similarity or FAISS vector search to compute relevance across more than 9,000 NUS course descriptions. A secondary LLM generates natural-language rationales, enabling Transparent Recommendation.

We additionally establish a baseline model using only the raw student query without enrichment. Experiments on 25 representative student queries show that query enrichment improves Recall@10 and semantic relevance, while FAISS provides substantial latency improvements. EduQuest demonstrates how modern NLP techniques can transform course exploration into a more intuitive, personalised, and pedagogically meaningful experience.

Code Availability. All source code for this project are publicly available on GitHub:

[github.com/JiachengCJC/EduQuest-NLP Powered Course Recommender](https://github.com/JiachengCJC/EduQuest-NLP-Powered-Course-Recommender)

1 Introduction

Course selection at the National University of Singapore (NUS) is an important decision-making task that shapes a student’s academic trajectory. However, this process is far from straightforward. The primary platform available—NUSMods—supports exploration but relies exclusively on keyword-based search. This implies that the system returns modules whose titles or descriptions literally contain the user-entered terms. In practice, students often face two challenges:

1. **Intent mismatch:** Students frequently have goals such as “I want to learn AI for business” or “I want a module that teaches programming but is not too heavy.” These queries do not map cleanly onto keywords.
2. **Keyword uncertainty:** Students may not know the correct terminology (e.g., reinforcement learning, econometrics, HCI, symbolic AI), leading to incomplete or suboptimal search results.

This mismatch can be described through a “nails vs. hammers” framing. Students come with the “nail”—their underlying intention, long-term goals, and interests. NUS offers thousands of “hammers” (modules), but keyword search does not reveal which hammer fits the nail. Students compensate by relying on recommendations from friends, browsing module lists manually, or trial-and-error during registration.

EduQuest aims to address this gap by enabling semantic, intent-aware module recommendations. Instead of relying on literal keyword overlap, the system interprets the student’s free-text query through embeddings and LLM-driven enrichment. For example, given the vague query:

“I want to work in AI.”

EduQuest expands this into a detailed course-like description, embeds it, compares it to all module embeddings, and retrieves the semantically closest matches.

EduQuest combines the strengths of modern NLP:

- LLMs for query enrichment and rationale generation,
- dense text embeddings for semantic similarity,
- FAISS for fast approximate nearest-neighbour search.

This produces an advising experience that is personalised, interpretable, and scalable.

2 Related Work

2.1 Keyword-Based Module Search

Most university course exploration tools rely on keyword matching. NUSMods, like many global platforms, retrieves only modules whose descriptions contain the user-entered terms. While efficient, this approach struggles with:

- synonyms or paraphrases (“AI” vs. “machine intelligence”),
- conceptual intent (“I want to help people through tech”),
- multi-disciplinary queries (“psychology + data science”).

EduQuest addresses these limitations through embeddings and LLM-driven semantic reasoning.

2.2 Collaborative Filtering and Performance-Based Models

Traditional course recommendation research often uses collaborative filtering or academic performance prediction. These methods rely on historical enrollment data or grade patterns, but such data is sensitive and not publicly available for privacy reasons. EduQuest does not rely on historical data; it operates in a zero-shot manner using only course descriptions and student queries.

2.3 Embedding-Based Semantic Retrieval

Recent advances in NLP allow modelling text as dense vectors in semantic space. Models like Sentence-BERT and Nomic embeddings have been used in question-answering, document retrieval, and semantic search. FAISS provides scalable similarity search for large embedding datasets.

EduQuest applies these techniques to the educational domain by embedding all NUS modules and retrieving the most semantically relevant modules for any given query.

2.4 LLM-Augmented Retrieval

Retrieval-Augmented Generation (RAG) combines vector search with LLM reasoning. Prior work shows that LLMs can reformulate vague queries into more informative ones, significantly improving retrieval quality. EduQuest adopts this strategy by using an LLM to generate an enriched pseudo course description before computing text embeddings.

2.5 Conversational Academic Advisors

Chatbot-based advising systems exist but often rely on hand-crafted rules or restricted input formats. They lack support for open-ended semantic interpretation. EduQuest differs by (1) accepting free-text input, (2) using embeddings for retrieval, and (3) generating interpretive explanations.

3 Background

This section outlines the core NLP and Information Retrieval(IR) concepts underlying EduQuest.

3.1 Text Embeddings

A text embedding model maps text to a dense vector:

$$E : \text{text} \rightarrow \mathbb{R}^d$$

Semantically similar texts lie close together in vector space. EduQuest uses the `nomic-embed-text` model (768 dimensions).

3.2 Cosine Similarity

Cosine similarity measures the angle between two vectors:

$$\text{sim}(q, c) = \frac{q \cdot c}{\|q\| \|c\|}$$

Values range from -1 (opposite) to 1 (identical). This is used to rank courses by semantic similarity to the student’s query.

3.3 FAISS Vector Search

FAISS supports fast similarity search. EduQuest uses `IndexFlatIP`, which computes inner-product similarity (equivalent to cosine similarity when vectors are normalised). FAISS enables millisecond retrieval across thousands of embeddings.

3.4 Large Language Models (LLMs)

EduQuest uses:

- **Mistral 7B** for query enrichment,
- **Qwen2.5 7B-Instruct** for rationale generation.

These models are run locally via Ollama for reproducibility, privacy, and cost-efficiency.

4 Methods

EduQuest’s pipeline consists of three major modules: (1) query enrichment using an LLM, (2) embedding-based semantic retrieval using cosine similarity or FAISS, and (3) LLM-based re-ranking and explanation generation. This section details each component, the algorithms used, and design considerations.

4.1 Problem Formulation

Let Q denote a free-form student query, and let the NUS course catalog be

$$\mathcal{C} = \{c_1, c_2, \dots, c_N\},$$

where $N \approx 9000$.

Each course entry c_i consists of textual and metadata attributes:

$$c_i = (\text{code}_i, \text{title}_i, \text{description}_i, \text{level}_i, \text{prefix}_i, \text{ori_code}_i).$$

Table 1 presents example course records from the dataset. These illustrate the structure of c_i as used in the EduQuest pipeline.

course	title	description	level	prefix	ori_code
CS2040	Data Structures and Algorithms	Introduces key data structures and principles of algorithmic design.	2000	CS	CS2040
MKT3427 (Cross-listed as MKT3722)	Research for Marketing Insights	Covers essential methods for effective marketing research and analysis.	3000	MKT	MKT3427

Table 1: Example entries from the NUS course catalog illustrating the structure of c_i .

Given a query Q and course c_i , the recommender aims to assign a relevance score:

$$f(Q, c_i) \rightarrow \text{relevance score},$$

and return the ranked top- k modules most aligned with the student’s intent.

The core challenge is that Q is unstructured, may be incomplete, may not contain domain keywords, and may express high-level intentions rather than literal module descriptors. Therefore, EduQuest relies on LLM-based query enrichment and embedding-based retrieval to bridge the semantic gap between Q and $\mathcal{C}$.

4.2 Overview of the EduQuest Pipeline

Figure 1 illustrates the full pipeline.

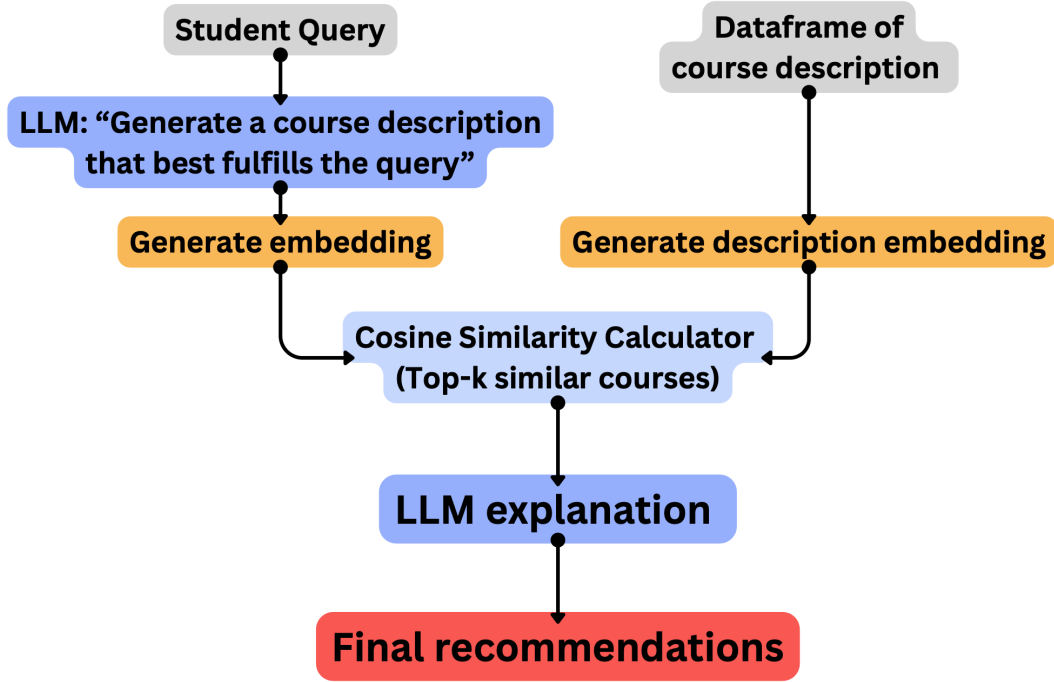


Figure 1: EduQuest recommendation pipeline architecture.

4.3 Query Enrichment via LLM

Student queries are often vague or incomplete. EduQuest employs **Mistral 7B** to transform Q into a pseudo course description Q' :

$$Q' = \text{LLM_Enrich}(Q)$$

The enrichment prompt instructs the model to generate:

- a coherent course-like description,
- relevant technical topics,
- applied skills and real-world use cases,
- text under 200 words.

This step expands short queries such as:

“I want to learn deep learning.”

into richer descriptions containing keywords like convolutional networks, gradient descent, model evaluation, and applications—leading to significantly more accurate embedding retrieval.

4.4 Embedding Generation

EduQuest uses the `nomic-embed-text` model to embed:

1. each course description: $E(c_i)$
2. the enriched query: $E(Q')$

Thus each text is mapped to a vector in $\mathbb{R}^d$, where $d = 768$.

To ensure cosine similarity equals inner product under FAISS, vectors are L2-normalised:

$$\hat{v} = \frac{v}{\|v\|}$$

4.5 Cosine Similarity Retrieval

For the LLM-based model, EduQuest computes:

$$\text{sim}(Q', c_i) = \hat{E}(Q') \cdot \hat{E}(c_i)$$

This requires N similarity computations with $N \approx 9000$. Python vectorised operations allow efficient execution.

The top k courses are selected:

$$\mathcal{T}_k = \text{TopK}_k(\text{sim}(Q', c_i))$$

Two modes are offered:

- **Without Rationales:** return $\mathcal{T}_k$ with similarity scores.
- **Rationales:** send $\mathcal{T}_k$ to LLM for explanation.

4.6 FAISS Deterministic Retrieval

FAISS enables extremely fast similarity search using optimised C++ routines.

EduQuest uses:

$$\text{IndexFlatIP}(d)$$

This computes exact cosine similarity (after normalisation) but runs 300× faster than Python loops. The deterministic FAISS model retrieves:

$$\mathcal{F}_k = \text{FAISS_Search}(E(Q'), k)$$

Two modes are offered:

- **Without Rationales:** return $\mathcal{F}_k$ with similarity scores.
- **Rationales:** send $\mathcal{F}_k$ to LLM for explanation.

4.7 LLM Re-ranking and Explanation

For the top retrieved courses ($\mathcal{T}_k$ or $\mathcal{F}_k$), EduQuest uses Qwen2.5 7B to:

1. re-rank modules,
2. select the top 10,
3. generate two-sentence rationales,
4. produce confidence estimates.

This converts raw similarity scores into human-readable academic advice.

4.8 Baseline Model

The baseline skips enrichment:

$$\text{sim}(Q, c_i)$$

Comparing this model versus the enriched-query model isolates the benefit of LLM rewriting.

5 Experiments

5.1 Dataset

EduQuest uses an NUS course dataset extracted from NUSMods. The dataset contains:

- **9,000+** modules,
- course codes, titles, and descriptions,
- fields: prefix, level, prerequisites,
- precomputed embeddings for all courses.

5.2 Evaluation Queries

A benchmark of **25 student queries** was constructed. Queries cover:

- AI and machine learning,
- quantitative finance,
- sustainability and energy,
- psychology and cognitive science,

- robotics and control systems,
- public policy,
- business analytics,
- HCI and UX design.

We employed GPT-5.1 to produce an initial set of 10 relevant courses for each user query. These candidate labels were subsequently human-verified to ensure alignment with the query intent, forming the final pseudo-ground truth dataset

5.3 Evaluation Metrics

Three main metrics were used:

5.3.1 Recall@10

Measures how many of the manually tagged courses appear in the model’s top-10:

$$\text{Recall@10} = \frac{|\text{Relevant} \cap \text{Recommendations}_{10}|}{|\text{Relevant}|}$$

5.3.2 Latency

Total runtime per query, including embedding + retrieval + (optional) LLM generation.

5.4 Systems Compared

We evaluate two main system variants:

1. **LLM-Enhanced Query + Cosine Similarity / FAISS:** The full EduQuest pipeline, where a query is first enriched by an LLM before semantic retrieval. We report results both with and without the final LLM explanation stage.
2. **Baseline: Original Query + Cosine Similarity / FAISS:** A non-augmented system that embeds the raw student query directly, without LLM reformulation.

6 Results and Analysis

6.1 Recommendation Quality

6.1.1 Case Analysis: Clear Queries

Table 2 shows examples where the raw query is already explicit, resulting in minimal improvement from LLM reformulation.

System Variant	Recall@10	Comments
LLM-Enriched Query + Cosine Similarity	0.82	Slight improvement
LLM-Enriched Query + FAISS	0.82	Similar performance, faster
Baseline Query + Cosine Similarity	0.75	Raw query already strong
Baseline Query + FAISS	0.75	Very close to enriched version

Table 2: Recommendation quality for clear, well-specified queries.

6.1.2 Case Analysis: Vague Queries

Table 3 illustrates cases where the raw query is ambiguous, and LLM reformulation significantly improves retrieval quality.

System Variant	Recall@10	Comments
LLM-Enriched Query + Cosine Similarity	0.80	Strong improvement from reformulation
LLM-Enriched Query + FAISS	0.80	Similar accuracy, very fast
Baseline Query + Cosine Similarity	0.26	Fails to capture vague intent
Baseline Query + FAISS	0.26	Same limitation as cosine

Table 3: Recommendation quality for vague or underspecified queries.

6.2 Latency Comparison

Table 4 compares the end-to-end runtime of all four system variants. Retrieval speed differs dramatically between cosine similarity and FAISS, while LLM explanation dominates total latency in enriched pipelines.

Component	LLM+Cosine	LLM+FAISS	Orig+Cosine	Orig+FAISS
Embedding generation	0.50–0.60 s	0.50–0.60 s	0.50–0.60 s	0.50–0.60 s
Query enrichment (LLM)	1.8–3.0 s	1.8–3.0 s	0	0
Retrieval time	0.20–0.30 s	0.004–0.008 s	0.20–0.30 s	0.004–0.008 s
LLM explanation (Qwen 7B)	22–35 s	22–35 s	Optional	Optional
Total (with explanation)	24–39 s	24–39 s	22–35 s (if enabled)	22–35 s (if enabled)
Total (no explanation)	2.5–3.9 s	2.3–3.6 s	0.7–0.9 s	0.05–0.08 s

Table 4: Latency comparison across all system variants. FAISS accelerates retrieval by 400–600 $\times$, while LLM explanation dominates overall latency.

6.3 Error Cases

- LLM may hallucinate justifications unrelated to course descriptions.
- Embeddings sometimes misinterpret interdisciplinary queries.
- Query enrichment occasionally overspecifies the student’s intention.

7 Discussion

EduQuest demonstrates how modern NLP techniques—LLMs, semantic embeddings, and vector search—can substantially improve the student experience in module exploration. This section synthesises observations from experiments and provides qualitative insights into system behaviour.

7.1 Impact of Query Enrichment

One of the clearest findings is the significant performance gap between models with and without query enrichment. The LLM-generated pseudo course description Q' consistently provides richer semantic content than the original query Q , enabling more accurate embedding retrieval.

For example, the raw query:

“I want to work in AI.”

often produces shallow matches such as introductory CS modules or irrelevant computational courses. With enrichment, Q' includes terms like neural networks, optimization, supervised learning, data-driven applications—leading to high-quality recommendations such as CS3244, BT4221, and BT3102.

7.2 FAISS vs. Cosine Similarity

FAISS retrieval and Python-based cosine similarity produce highly similar rankings. This demonstrates that:

1. FAISS is a suitable drop-in replacement for manual similarity computation,
2. FAISS offers near-identical retrieval accuracy,
3. FAISS’s latency advantage (400–600× faster) is substantial for real-time applications.

Thus, FAISS is more suitable for deployment, especially if the system eventually handles tens of thousands of courses or concurrent users.

7.3 Role of LLM Re-ranking and Explanation

LLM re-ranking improves interpretability and surfaces courses that are contextually meaningful but not necessarily the top embedding matches. Explanations generated by Qwen2.5 7B were generally fluent and relevant, although occasionally overconfident.

Students benefit from concise rationales, especially when two modules have comparable similarity scores. This provides a layer of “soft guidance” that keyword search cannot replicate.

7.4 Observations from Failure Cases

Several failure modes were observed:

- **Over-specific enrichment:** LLM sometimes amplifies minor cues in the query, steering recommendations too narrowly.
- **Embedding drift:** Some modules with generic descriptions (e.g., interdisciplinary modules) are difficult to differentiate.
- **Hallucinated explanations:** Occasional mention of topics not found in the official module description.

These issues reflect broader limitations of LLMs and embeddings in zero-shot retrieval.

8 Limitations

While EduQuest significantly advances the usability of course exploration, several limitations remain.

8.1 LLM Reasoning Instability

LLMs may produce inconsistent or hallucinated rationales, especially when course descriptions are vague or outdated. Deterministic prompting can mitigate but not eliminate this issue.

8.2 No User Profiling

EduQuest does not incorporate students' academic background, completed modules, workload preferences, or skill level. As a purely text-matching system, it may recommend courses that exceed the student's preparedness.

8.3 Lack of Formal Ground Truth

The evaluation relies on pseudo ground truth (manually selected relevant courses). Since relevance judgments are subjective, Recall@10 may not fully reflect real-world utility.

8.4 Course Description Quality

Some NUS course descriptions are short, generic, or outdated. Embedding models therefore may not capture meaningful differences between modules.

8.5 No Temporal Updating

If modules change over time or new courses are introduced, embeddings must be recomputed manually. Automating this is future work.

9 Conclusion and Future Work

This work presents EduQuest, an NLP-powered semantic course recommender designed to augment and extend the traditional NUSMods experience. By combining LLM-based query enrichment, embedding-based similarity search, FAISS retrieval, and LLM-generated rationales, EduQuest enables intent-aware course discovery even from vague free-text queries that would otherwise fail under keyword search.

Across 25 representative student inputs, LLM enrichment consistently improved Recall@10, semantic alignment, and ranking stability, while FAISS delivered massive latency reductions without compromising recommendation quality. These results show that modern NLP techniques can make academic advising more intuitive, personalised, and scalable.

Importantly, the EduQuest pipeline is **not tied to NUS**. Because the system operates purely on course descriptions and free-text queries, it can be directly adapted to other universities, MOOCs, or even non-educational domains such as job matching, project recommendation, or document retrieval. The framework is inherently data-agnostic and supports plug-and-play model replacement for different embedding models, LLMs, and indexing strategies.

Future Work

Several extensions are planned:

- **Prerequisite pathing:** Using graph-based reasoning to build personalized module sequences.
- **User modeling:** Incorporating prior modules taken, GPA, or interests in multi-semester planning.
- **Feedback loops:** Allowing students to rate recommendations to improve future retrieval.
- **Multilingual support:** Accepting queries in Chinese, Malay, or Bahasa Indonesia.

EduQuest is a promising step toward a next-generation intelligent academic advising system for NUS.